



Statistical Sampling and Data Collection

• Mark Scheme

Statistics Chapter 1: Statistical Sampling and Data Collection. M = method mark A = accuracy mark (depends on M)
B = independent mark ft = follow through

Section A – Populations, samples and sampling frames

- 1 (a) Every member of the group being studied *B1*. (b) A list of all the members of the population that the sample can be chosen from *B1*.
- 2 Testing is destructive (a tested battery cannot be sold) *B1*; a census would take too long / cost too much *B1*.
- 3 (a) E.g. the electoral register for the town *B1*. (b) E.g. it may be out of date, or miss residents who are not registered *B1*. (c) There are too many residents, so a census would be too costly and slow *B1*.
- 4 Only students who eat school meals are asked *B1*; students who bring packed lunches (who may dislike the meals) are excluded *B1*; so the results are likely to be biased towards a more positive view *B1*.
- 5(a) All 2000 members of the gym *B1*.
- 5(b) Members who rarely visit are missing from the frame *B1*; they may be less satisfied, so satisfaction would be over-estimated *B1*.
- 5(c) The full membership list of all 2000 members *B1*; every member can be chosen, so no group is excluded *B1*.
- 5(d) Advantage: gives the views of every member, so fully representative *B1*. Disadvantage: time-consuming/costly, and many may not respond *B1*.

Section B – Sampling techniques

- 6 Number the students 001 to 400 *B1*; use a random number generator to pick 30 different numbers, ignoring repeats and numbers above 400 *B1*.
- 7 (a) $k = 500 \div 20 = 25$ *B1*; choose a random start between 1 and 25, then take every 25th customer *B1*. (b) E.g. bias if the list has a pattern matching the interval *B1*.
- 8 Sampling fraction $\frac{50}{400}$ *M1*; boys $240 \times \frac{50}{400} = 30$ *A1*; girls 20 *A1*.
- 9 (a) $120 \times \frac{40}{250} = 19.2 \rightarrow 19$, $85 \times \frac{40}{250} = 13.6 \rightarrow 14$, $45 \times \frac{40}{250} = 7.2 \rightarrow 7$ *M1 A1 A1*. (b) Number the 120 employees and choose 19 by simple random sampling *B1*.
- 10(a) Stratified: members chosen *at random* within each group; quota: the interviewer chooses members to fill each group, not at random *B1 B1*.
- 10(b) Advantage: quick/cheap, no sampling frame needed *B1*. Disadvantage: not random, so interviewer bias is possible *B1*.
- 10(c) Total 300 *B1*; juniors $84 \times \frac{25}{300} = 7$, adults 13 *M1*; seniors 5 *A1*.
- 10(d) Only members who use the café would be asked, so it would not represent all members *B1*.

Section C – Types of data

- 11 (a) Quantitative, discrete *B1*. (b) Qualitative *B1*.
- 12 (a) Continuous *B1*; (b) continuous *B1*; (c) discrete *B1*.
- 13 Amira: primary, she collected it herself for her study *B1*. Ben: secondary, collected by someone else for another purpose *B1*.
- 14 Shoe sizes can only take certain fixed values (no size between 7 and $7\frac{1}{2}$) *B1 B1*; time can take any value in a range, and rounding to 0.1 s is only how it is recorded *B1*.



15(a) Cloud cover is discrete: it only takes whole-number values 0 to 8 *B1*. Rainfall is continuous: it can take any value in a range *B1*.

15(b) Treat “tr” as 0 *B1*, since the amount is less than 0.05 mm, which is 0.0 to 1 decimal place *B1*.

15(c) Only one month of one year is used *B1*; rainfall varies with season and from year to year (e.g. climate change) *B1*.

15(d) Leave it out, so the mean is found from the remaining 3 days *B1*.

Roar-level challenge

16 Sampling fraction $\frac{60}{1500} = 0.04$ *M1*; 12.8, 12.4, 12, 11.6, 11.2 *A1*; rounded: 13, 12, 12, 12, 11 (total 60) *A1*; each year group is represented in proportion to its size, so the sample reflects the whole school *B1*.